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In [4]: import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression

# Dataset: marks vs result
X = np.array([[30], [35], [40], [45], [50], [55], [60], [65]])
y = np.array([0, 0, 0, 0, 1, 1, 1, 1]) # 0 = Fail, 1 = Pass

# Create and train model
model = LogisticRegression()
model.fit(X, y)

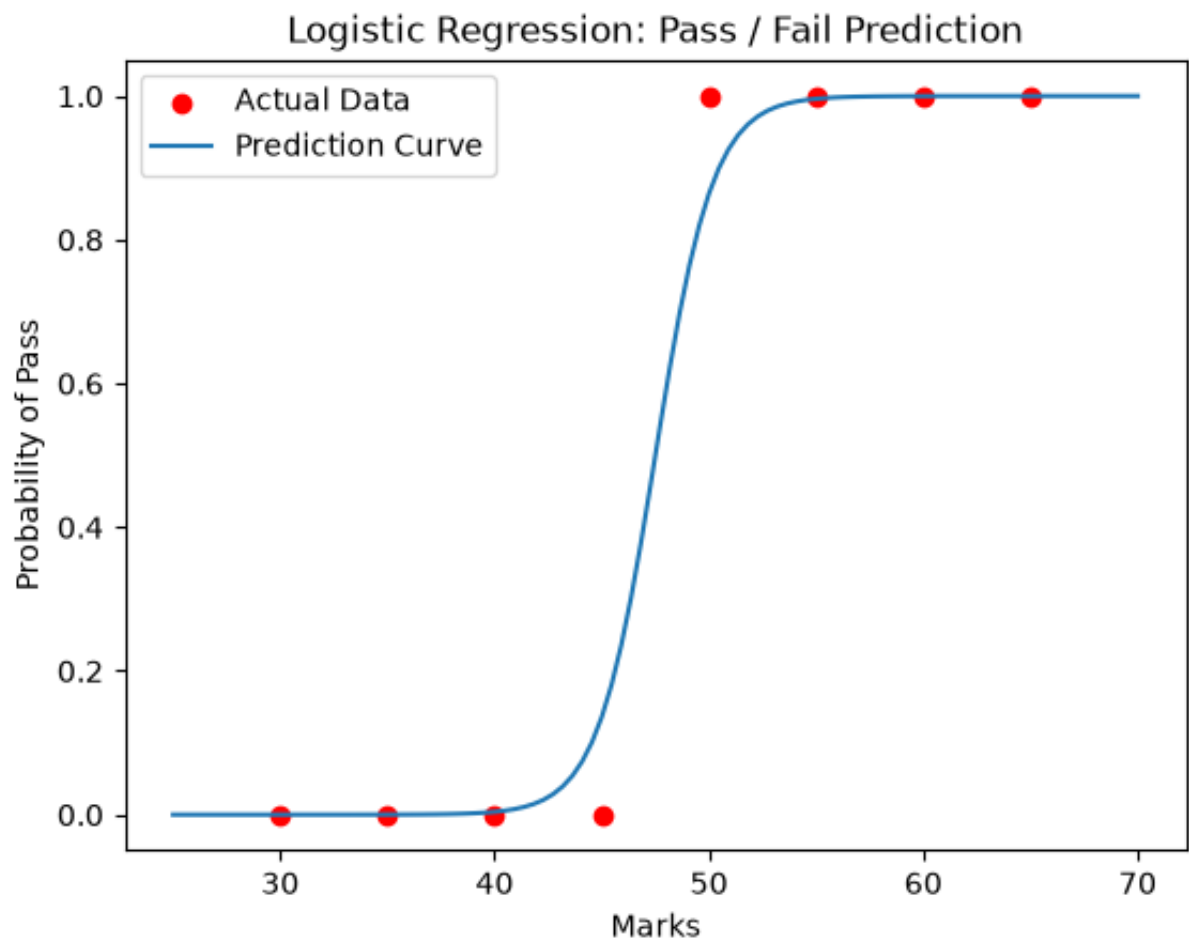
# Predict for a new student
marks = [[48]]
result = model.predict(marks)
print("Prediction (0=Fail, 1=Pass):", result)

# Plot data points
plt.scatter(X, y, color='red', label='Actual Data')

# Plot logistic regression curve
X_test = np.linspace(25, 70, 100).reshape(-1, 1)
plt.plot(X_test, model.predict_proba(X_test)[:,1], label='Prediction Curve')

plt.xlabel("Marks")
plt.ylabel("Probability of Pass")
plt.title("Logistic Regression: Pass / Fail Prediction")
plt.legend()
plt.show()
```

Prediction (0=Fail, 1=Pass): [1]



In []: